

Phase 1 - Week 4 - Lesson Day 2

Hypothesis Testing

Today's Goal

Today we learn how to check if a difference in data is **real**, or just **random chance**. This is called hypothesis testing. It is used everywhere in data science, business, and medicine.

- Understand null and alternative hypothesis
- Understand what a p-value means
- Run a simple t-test

1. Null Hypothesis and Alternative Hypothesis

Simple definition: Before you test anything, you write down two possible answers.

Null hypothesis (H0): "There is NO real difference. Any difference we see is just random chance."

Alternative hypothesis (H1): "There IS a real difference."

Example: you want to know if Students who sleep more score higher on tests. Your null hypothesis is "sleep does not affect test scores." Your alternative hypothesis is "sleep DOES affect test scores." The test will help you decide which one to believe.

2. What Is a P-Value?

Simple definition: The p-value is a number between 0 and 1. It tells you how likely it is that the difference you see happened by random chance, if the null hypothesis is true.

Small p-value (usually below 0.05): the difference is unlikely to be random chance. We reject the null hypothesis. We believe there IS a real difference.

Large p-value (above 0.05): the difference could easily be random chance. We do NOT reject the null hypothesis. We do not have enough proof of a real difference.

0.05 is just a common rule that many people use. It means "only a 5% chance this happened randomly." It's not a perfect law, just a common line people use to decide.

3. Running a T-Test

Simple definition: A t-test compares the averages of two groups, and tells you the p-value - how likely it is that the difference between them is just random chance.

```
from scipy import stats

# Test scores from two different classes
class_a = [70, 72, 68, 75, 71, 69]
class_b = [80, 85, 78, 82, 88, 79]

t_stat, p_value = stats.ttest_ind(class_a, class_b)

print("p-value:", p_value)

if p_value < 0.05:
    print("The difference is likely REAL (not random chance)")
else:
    print("The difference could be random chance")
```

stats.ttest_ind() runs the t-test for you. You give it two groups of numbers, and it gives back a p-value. You don't need to know the complex math behind it - just know how to read the result.

Today's Practice

Open **day-2-hypothesis-testing.ipynb**. All answers are already solved with simple code and clear comments.

Practice Questions (4 total - all solved)

1. Write the hypotheses

For a test comparing two study methods, write the null hypothesis and the alternative hypothesis in plain words (as a comment).

2. Run a t-test

Compare two small groups of made-up exam scores using a t-test, and print the p-value.

3. Interpret the result

Using your result from Question 2, write a comment saying if the difference is likely real or just random chance.

4. Try a bigger difference

Change the numbers so the two groups are very different from each other, run the t-test again, and compare the new p-value to Question 2's.

Once you and Halimo finish reviewing today's practice, move to Lesson Day 3 (Confidence Intervals).